

Settlement Pivot Edge Research

Proprietary market-structure edge, live scanner, and historical validation on XAUUSD

1 Research idea: the edge is not the entry

I developed a proprietary market-structure edge for conditions in which a strong directional impulse is followed by a structured correction and break. The edge identifies a context where at least a partial settlement / correction or reversal becomes more likely; it does not by itself define an entry. Execution requires separate evidence of momentum change on a lower fractal / timeframe.



Settlement Pivot edge scanner

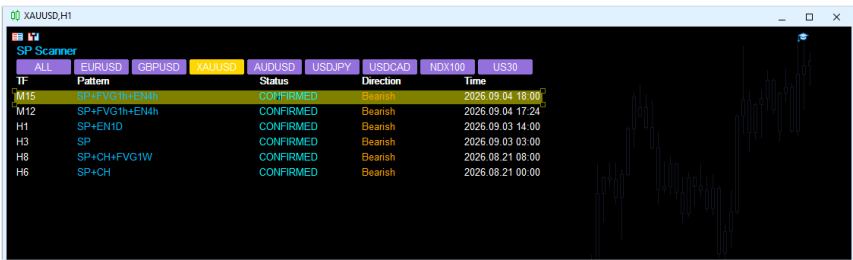


Figure 1. Scanner dashboard reporting multi-timeframe Settlement Pivot edge candidates on XAUUSD.

2 What I built

Edge + scanner: the scanner operationalizes the proprietary higher-level structure and reports bullish or bearish edge candidates; the edge is not a standalone entry trigger.
Lower-fractal execution: an actual entry requires separate evidence that momentum is changing on a lower fractal / timeframe; different confirmation structures can be used.
Standardized validation proxy: for historical testing, I used a deliberately simple close-based break associated with an engulfing-type confirmation candle, then exported verified events for matched-baseline analysis.

Signal and confirmation example

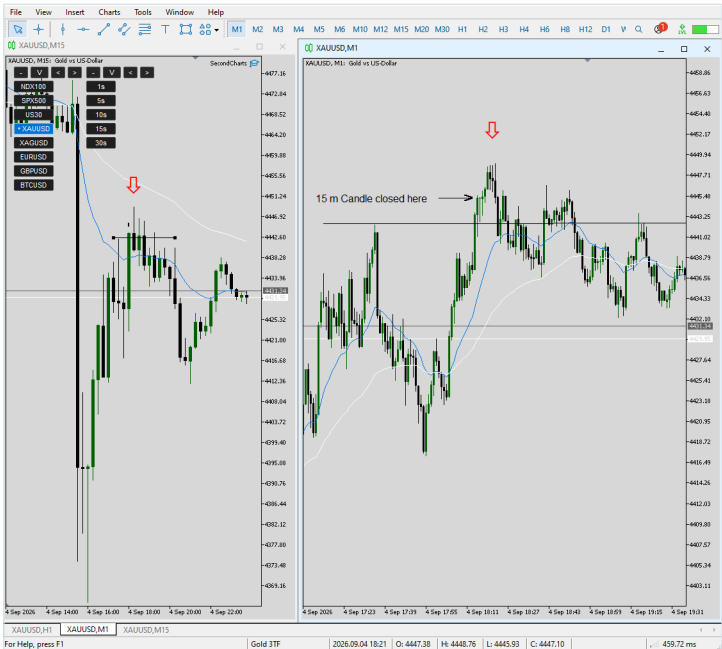


Figure 2. Higher-timeframe edge context (left) and lower-fractal context (right). The close-based confirmation shown here is the standardized proxy used for historical testing; other lower-fractal momentum-shift confirmations may also be used for execution.

Confidentiality Signal logic, thresholds, and parameterization are intentionally omitted. The work sample documents the research process, tooling, and aggregate validation evidence without making the strategy reconstructable.

Historical Validation

Matched non-signal controls using a standardized lower-fractal confirmation proxy

Sample
XAUUSD, 2021-01-01 to 2026-08-31. M30 and H1 emphasized.

Signal observations
234 total; 220 evaluable. 14 same-bar ambiguous observations excluded from the primary rate.

Matched controls
1,170 total; 1,107 evaluable. Five non-signal controls per signal.

Primary outcome
+1.0 ATR in the predicted direction before -0.5 ATR invalidation within 30 bars.

Selected matched-control results

Segment	Signal N T / E	Control N T / E	Signal success	Matched baseline	Lift (pp)
Overall M30+H1	234 / 220	1,170 / 1,107	40.9%	35.7%	+5.2 pp
M30 overall	166 / 154	830 / 788	41.6%	36.9%	+4.6 pp
M30 Bullish	69 / 64	345 / 332	45.3%	37.3%	+8.0 pp
H1 overall	68 / 66	340 / 319	39.4%	32.6%	+6.8 pp
H1 Bearish	38 / 36	190 / 178	41.7%	29.2%	+12.5 pp

T / E = Total / Evaluable. Primary rates exclude AMBIGUOUS_SAME_BAR observations.

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Interpretation

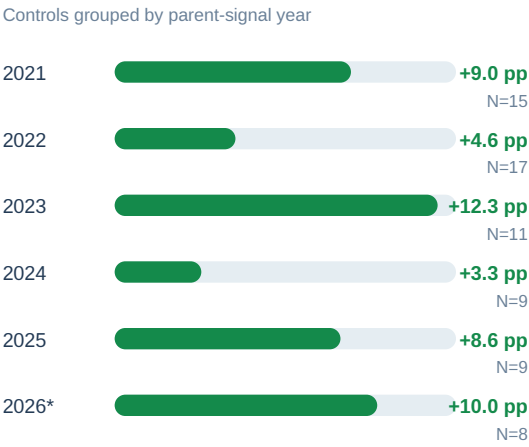
Overall: 40.9% evaluable-event success versus 35.7% matched baseline, a +5.2 pp historical lift.

Conservative ambiguity check: counting every same-bar ambiguous case as a failure gives 38.5% versus 33.8%, leaving approximately +4.7 pp lift.

H1 Bearish shows the largest reported lift (+12.5 pp), but the subgroup is smaller.

These results are best described as historical evidence of conditional lift, not proof of persistent future alpha.

M30 Bullish - yearly lift vs matched baseline



Method note

Primary success rates exclude same-bar cases where target and invalidation ordering cannot be resolved from bar data. The same outcome test is applied to signals and controls. Annual controls remain linked to the year of their parent signal.

No-Look-Ahead Manual Backtesting Engine

Operator-driven MT5 backtesting with synchronized multi-chart and configurable second-based replay

1 Manual backtesting gap and core innovation

I designed this engine for manual, operator-driven backtesting in MT5. It combines no-look-ahead historical replay, synchronized multi-chart review, and user-configurable second-based periods in one environment. At the time of development, I had not found an MT5 tool that combined these capabilities in one integrated manual-backtesting workflow.

Manual backtesting

Step controls and variable-speed replay for operator-driven testing.

Configurable second periods

The user selects the S<number> intervals needed for each study.

Synchronized charts

Leader and follower charts share the same simulated replay clock.

2 Backtesting design principle

A single simulated clock defines what the operator can see. Only historical data up to the current replay timestamp are exposed, limiting future-bar leakage. Native MT5 periods and user-defined second periods are generated to the same cutoff, while synchronized follower charts receive the same replay state.

Manual backtesting across synchronized timeframes

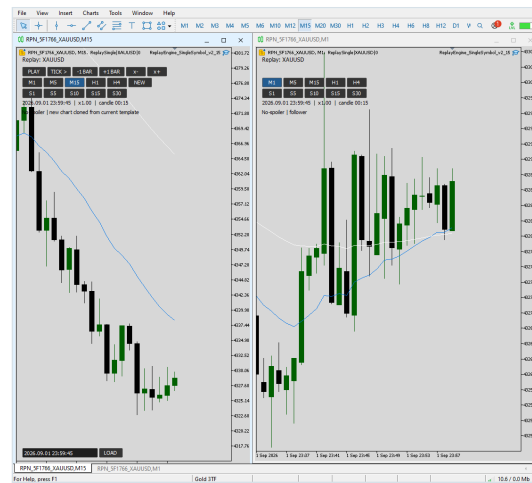
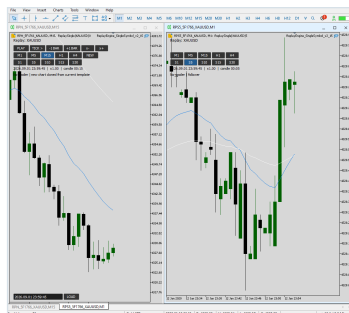
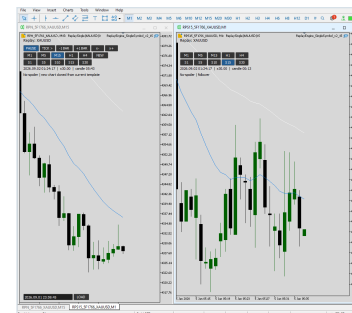


Figure 1. M15 and M1 in synchronized manual replay. Future data remain withheld until the replay clock advances.



M15 + S5

Five-second follower view for sub-minute inspection.



M15 + S15

15-second replay with variable-speed and step controls.

Integrated Live Market Research Toolkit

A custom MT5 platform-extension layer for persistent, second-based, multi-chart research

SymbolJump

In MT5, repeated symbol changes can reload chart/template context and disturb the working layout. SymbolJump provides one-click symbol switching while preserving the research workspace, chart organization, and template context.

SecondCharts

MT5 has no native second-based chart periods. My design uses configurable seconds slots: the user can choose the second intervals needed for the workflow (1s / 5s / 10s / 15s / 30s are examples). Switching between the chosen seconds views - and returning to the standard chart - is immediate, without forcing a separate-chart workflow.

SyncedCharts

Time, crosshair, and drawing-object synchronization across linked charts, including standard and second-based views. This reduces manual alignment work and keeps multi-timeframe annotations spatially consistent.

Platform-extension concept

The goal was not to replace MetaTrader, but to extend a rigid, friction-heavy default workflow into a more responsive research environment. The key design idea is the integration: persistent chart layouts, fast cross-symbol switching, native-like access to user-configurable second-based periods, and synchronized multi-chart analysis in one workflow.

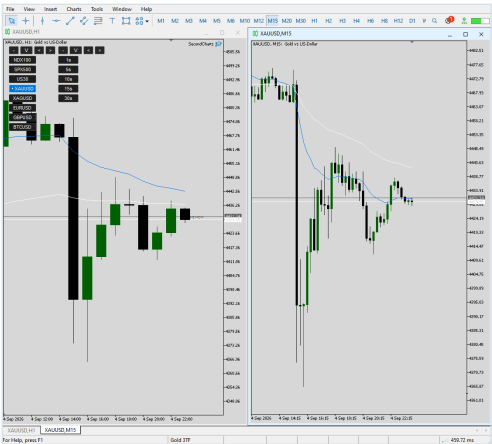


Figure 1. XAUUSD workspace

Standard and second-based controls within the same multi-chart research layout.

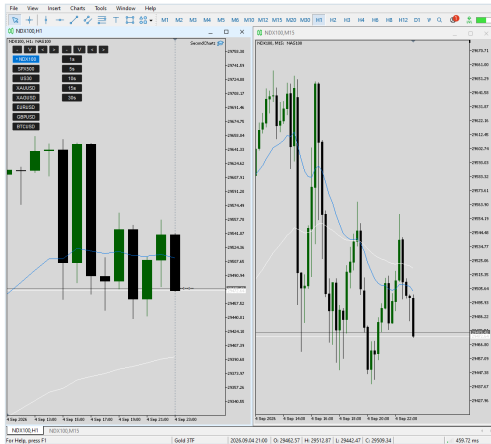


Figure 2. SymbolJump

NDX100 loaded with the chart organization preserved after a one-click symbol switch.



Figure 3. SecondCharts + standard chart

A second-based view inspected beside a standard timeframe chart.

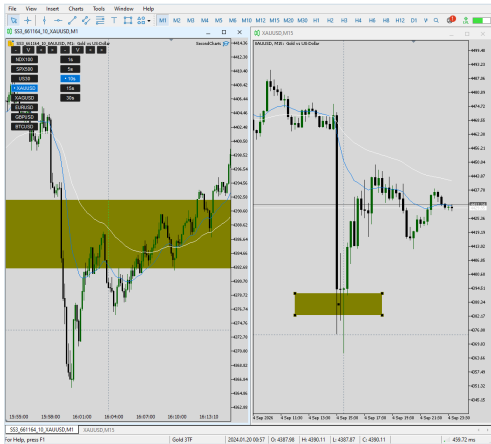


Figure 4. SyncedCharts

Time and rectangle-object alignment across linked 10-second and 15-minute views.

Design contribution

At the time of development, I had not encountered an MT5 workflow using this exact combination. Existing second-period tools I found generally centered on creating a separate chart or exposing a selected custom period; this design instead lets the user configure the seconds intervals, switch immediately among them, return to the standard view with one click, preserve workspace structure, and keep analysis synchronized across charts.